

ProspectIQ

Tools & Technologies Reference

XLVentures.AI Hackathon 2025

1. PLATFORM OVERVIEW

ProspectIQ is a multi-agent AI platform for automated B2B customer discovery. It is built on a modern Python stack combining a large language model (LLM) inference API, a real-time web search API, a multi-agent orchestration framework, a REST API backend, and an interactive frontend dashboard. All components are open-source or free-tier SaaS.

2. LLM & INFERENCE

Groq API

Role	LLM inference provider — hosts and serves the language model
Free Tier	500,000 tokens/day (llama-3.1-8b-instant)
Why chosen	Fastest open-source inference available; free tier sufficient for hackathon workloads
Website	https://console.groq.com

LLaMA 3.1 8B Instant

Role	Core language model powering all 7 AI agents
Parameters	8 billion
Context	128k tokens
Provider	Meta (served via Groq)
Why chosen	Fast, capable, and free; ideal for structured output generation across multiple agent calls

3. AGENT ORCHESTRATION

LangGraph

Version	0.4.8
Role	Multi-agent pipeline orchestration using a directed StateGraph

Pattern	Sequential 7-node graph; each node is an agent function that reads/writes shared state
State type	ProspectIQState (TypedDict) — typed shared memory across all agents
Why chosen	Purpose-built for stateful multi-agent workflows; clean separation of agent logic
Website	https://langchain-ai.github.io/langgraph/

LangChain

Version	0.3.25
Role	LLM integration layer; provides ChatGroq wrapper
Usage	llm.invoke(prompt) pattern used in every agent
Website	https://python.langchain.com/

4. WEB SEARCH

Tavily API

Role	Real-time web search — finds actual companies, decision makers, and contact details
Usage	Web Search Agent (parallel), Decision Maker Agent (per company), Enrichment Agent (per contact)
Search mode	basic depth, max 3–4 results per query
Free Tier	1,000 searches/month
Why chosen	LLM-optimised search API; returns clean snippets ready for prompt injection
Website	https://tavily.com

5. BACKEND

FastAPI

Version	0.115
Role	REST API backend exposing the agent pipeline to the frontend
Endpoints	POST /api/analyze, POST /api/approve, POST /api/report, GET /api/sessions, GET /api/sessions/{id}
CORS	Fully open (allow_origins=["*"]) for local development
Server	Uvicorn 0.34.3 (ASGI)
Why chosen	Fast, async, auto-generates OpenAPI docs; minimal boilerplate
Website	https://fastapi.tiangolo.com

Pydantic

Version	2.11.4
Role	Request/response validation for all API endpoints
Why chosen	Native to FastAPI; enforces typed contracts between frontend and backend

SQLAlchemy + SQLite

Role	Persistent storage for discovery sessions via DiscoverySession ORM model
DB file	prospectiq.db — auto-created on startup via init_db()
Usage	save_session(), get_all_sessions(), get_session_by_id(), update_session_status()

6. FRONTEND

Streamlit

Version	1.45
Role	Interactive web UI — input form, 6-tab results dashboard, human approval, past sessions sidebar
Port	8501
Components	upload_section.py, results_section.py, approval_section.py
Why chosen	Rapid UI development in pure Python; no HTML/JS required
Website	https://streamlit.io

7. PDF REPORT GENERATION

PyMuPDF (fitz)

Version	1.25.5
Role	Generates the 8-page professional PDF discovery report
Output	prospectiq_report.pdf — branded, downloadable from the Streamlit UI
Why chosen	Fast, precise PDF layout control with low-level drawing primitives
Website	https://pymupdf.readthedocs.io

8. INFRASTRUCTURE & CONFIGURATION

Python	3.13 — core language for all agents, backend, and frontend
python-dotenv	Loads API keys from .env file at startup
Keys managed	GROQ_API_KEY, TAVILY_API_KEY, GROQ_MODEL
Config module	config/settings.py — also declares ANTHROPIC_API_KEY and GEMINI_API_KEY for future use

9. SUMMARY TABLE

Technology	Version	Layer	Purpose
Groq API	—	LLM	LLM inference hosting
LLaMA 3.1 8B	—	LLM	Language model for all 7 agents
LangGraph	0.4.8	Agents	Multi-agent pipeline orchestration
LangChain	0.3.25	Agents	LLM integration layer (ChatGroq)
Tavily API	—	Search	Real-time web search
FastAPI	0.115	Backend	REST API (5 endpoints)
Uvicorn	0.34.3	Backend	ASGI server
Pydantic	2.11.4	Backend	Data validation
SQLAlchemy	—	Backend	ORM for session persistence
SQLite	—	Backend	Local database (prospectiq.db)
Streamlit	1.45	Frontend	Interactive UI + past sessions sidebar
PyMuPDF (fitz)	1.25.5	Output	8-page PDF report generation
Python	3.13	Core	Primary language
python-dotenv	—	Config	Environment variable management

Built for XLVentures.AI Hackathon 2025 · Powered by LangGraph, FastAPI, and Streamlit